

ECONOMIC POLICY UNCERTAINTY AND CARBON EMISSIONS IN EMERGING MARKETS

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ABSTRACT

This paper empirically examines the impact of economic policy uncertainty (EPU) on carbon emissions in 13 emerging market economies (EMEs), using panel local projections over the period 1990 to 2023. The results indicate that rising EPU significantly increases carbon emissions. However, the impact varies across economic and institutional characteristics. Specifically, EMEs with lower initial carbon emissions, higher energy intensity, carbon pricing mechanisms, greater financial development, higher trade openness, stronger political stability, greater renewable energy reliance, and higher research and development intensity exhibit a weaker response to EPU. Moreover, high-income EMEs experience a muted response in emissions, whereas lower-income EMEs face a more pronounced impact. Additionally, EMEs with low climate vulnerability see a stronger positive response in emissions compared to their high-vulnerability counterparts. These findings highlight the critical role of institutional and structural factors in shaping the emissions response to EPU, offering important policy insights for mitigating environmental risks in uncertain economic conditions.

Keywords: economic policy uncertainty, carbon emissions, emerging market economies

JEL codes: O13, Q54, Q58

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I. INTRODUCTION

Economic policy uncertainty (EPU) has become an increasingly significant concern in recent years, particularly in emerging market economies (EMEs), where fluctuations in policy uncertainty can substantially affect macroeconomic outcomes (Baker, Bloom, and Davis 2016). While much of the existing literature has focused on the effects of EPU on economic growth, investment, and financial markets, its impact on environmental outcomes, particularly carbon emissions, remains underexplored. Given that EMEs now account for around 60% of global carbon emissions (based on data from the 2024 Global Carbon Project), it is crucial to understand how EPU influences carbon emissions in these economies given that they play a central role in global efforts to mitigate climate change.

Our paper is motivated by the observation that EPU can affect a wide range of economic decisions and outcomes. Given the connection between economic activity, energy use, and environmental performance, EPU may be related to carbon emissions, a fact that warrants empirical examination. Broadly, policy uncertainty influences firms' financial decisions, investment strategies, and consumer behavior, all of which affect energy consumption and carbon emissions. Despite growing attention to the relationship between EPU and carbon emissions, there remains limited empirical evidence on the dynamic impacts of EPU on emissions, particularly from the perspective of EMEs. This paper aims to fill this gap in the literature by employing panel local projections (Jordà 2005) to estimate the impulse responses of carbon emissions to changes in EPU across 13 EMEs from 1990 to 2023, while also identifying the institutional and structural factors that shape the impact of EPU. By examining how economic and institutional characteristics affect the emissions response, this paper seeks to offer more granular insights into the relationship between policy uncertainty and environmental outcomes.

Conceptually, EPU can influence carbon emissions through several mechanisms. Earlier studies have shown that heightened policy uncertainty discourages investment, delays production decisions, and reduces aggregate economic activity (Bloom, 2009). Under this mechanism, lower output and energy demand would be expected to reduce carbon emissions. However, this perspective assumes that uncertainty primarily affects the volume of economic activity as opposed to altering the composition of investment or the pace of economic adjustment.

Policy uncertainty could also slow the transition toward a low-carbon economy through its effects on the composition and timing of investment rather than solely its overall level. Investments in renewable energy, energy efficiency, and clean technologies typically involve high upfront costs, long payback periods, and dependence on stable policy environments. Under heightened

uncertainty, firms may postpone replacing existing carbon-intensive capital or defer investments in cleaner technologies, prolonging reliance on more emissions-intensive production processes (e.g. Yu et al., 2025). Governments could also delay environmental regulations or public investment in green infrastructure, further slowing structural decarbonization. While the empirical analysis in this paper does not directly disentangle the contractionary effect on economic activity from the effect of slower structural adjustment, the results should be interpreted as reduced-form evidence of the net impact of EPU on emissions.

Our baseline results indicate that carbon emissions increase significantly following a rise in EPU. Specifically, a 1-standard deviation increase in EPU leads to a maximum increase of 0.03% in carbon emissions. This finding aligns with the broader literature, which has identified a positive relationship between EPU and carbon emissions (e.g., Benlemlih and Yavaş 2024, Chiou et al. 2025). These baseline estimates serve as a key benchmark for analyzing heterogeneous effects conditioned on various economic and institutional factors. Additionally, we find our results are robust to a range of sensitivity checks, including alternative model specifications and the inclusion of additional controls.

Importantly, we observe significant heterogeneity in the response to EPU across different economies, with the magnitude and persistence of the impact varying depending on each economy's structural and institutional characteristics. Specifically, economies with lower initial carbon emissions, higher income, higher energy intensity, carbon pricing mechanisms, greater financial development, higher trade openness, stronger political stability, greater reliance on renewable energy, and higher research and development (R&D) intensity tend to exhibit a weaker response to EPU. This suggests these economies may be better equipped to manage the adverse environmental impacts of EPU. Furthermore, we find the effects of EPU on carbon emissions also vary by income level: high-income EMEs experience a weaker response in emissions, whereas low-income EMEs face a more persistent and pronounced effect. This highlights the vulnerability of lower-income economies to policy-induced environmental changes. In addition, we observe that EMEs with lower climate vulnerability show a stronger positive response to EPU, in contrast to those with higher climate vulnerability, which may face more severe challenges in balancing economic and environmental priorities.

Our findings underscore the importance of considering structural and institutional factors when assessing the effects of EPU on environmental outcomes. The heterogeneity observed in the responses to EPU emphasizes the critical role of these factors in shaping the emissions trajectory in EMEs. As such, policymakers in EMEs need to account for their specific institutional

and structural characteristics when designing policies to mitigate the adverse effects of EPU on carbon emissions. The remainder of this paper is structured as follows. Section II discusses the related literature. Section III describes the data and empirical methodology. Section IV presents the baseline results with robustness checks. Section V presents the results of the extended analysis. Section VI concludes.

II. RELATED LITERATURE

Our paper is mainly related to the strand of the literature on the economic consequences of EPU. EPU can be broadly defined as “the economic risk associated with ill-defined future governmental policies and regulatory frameworks” (Al-Thaqeeb and Algharabali 2019). Such uncertainty affects economic decision-making at the national, corporate, and individual levels. Businesses often adopt a “wait-and-see” approach during periods of elevated EPU, delaying investment and spending decisions. It follows that our paper is linked to the implications of real options theory for investment, which argues that firms tend to delay investment when uncertainty is high, preferring to wait for clearer information (Dixit and Pindyck 1994, Bloom 2009). As investment for cleaner energy sources such as renewables is capital-intensive and requires substantial upfront outlays, this may be postponed when uncertainty is high, leading to higher carbon emissions, at least in the short run. Related theoretical work suggests higher uncertainty also leads to a higher cost of capital and tighter credit conditions, which hampers investment (Bernanke 1983; Gilchrist, Sim, and Zakrajšek 2014).

Several empirical studies have typically found adverse impacts of EPU on sentiment and broader economic activity (Bloom 2014; Baker, Bloom, and Davis 2016; Li, Su, and Wang 2022). High uncertainty leads households and businesses to adopt risk-averse behaviors, thereby reducing consumption, slowing economic growth, and increasing unemployment (Caggiano, Castelnuovo, and Figueres 2017; Al-Thaqeeb, Algharabali, and Alabdulghafour 2020). While substantial work already exists on the impact of EPU on economic activity and prospects, studies examining the effects of EPU on longer-term development goals, such as green economy transition and reducing carbon emissions, remain scarce. Our paper focuses on this aspect, notably from an emerging economy perspective.

A further strand of the related literature exists on the foundations of economy-specific EPU and the role of global shocks, whereby major global financial crises and geopolitical pressures have tended to be associated with heightened economic uncertainty. These would include, for example, the 2008 global financial crisis, the coronavirus disease pandemic, Russia’s war in

Ukraine, and United States global trade tariffs in 2025. Policy responses by national authorities in periods of elevated global uncertainty are fraught with difficulty, with economy-specific EPU rising as a result (Rodrik 1991, Aizenman and Marion 1993, Bloom 2009, 2014). EPU can also be the result of idiosyncratic factors and economy-specific financial volatility or political instability (Ozturk and Sheng 2018).

Our paper also relates to the literature on carbon pricing as a mechanism for achieving lower emissions, which has grown in recent years. Following the 2015 Paris Climate Agreement, many economies implemented regulatory measures to curb carbon emissions, with carbon pricing emerging as one of the most effective market-based instruments. Carbon pricing not only reduces emissions but also promotes clean technology investments. Best, Burke, and Jotzo (2020) provide empirical evidence that economies with carbon pricing policies have experienced a 2-percentage point reduction in the annual growth rate of carbon dioxide (CO₂) emissions compared to those without such policies. This underscores the importance of regulatory frameworks in shaping emission trajectories.

However, evaluating the impact of carbon pricing on emission reductions remains complex (Sumner, Bird, and Dobos 2011; Meckling, Sterner, and Wagner 2017; Haites 2018). This complexity arises from variations in coverage and stringency across jurisdictions, as well as interactions with other climate policies, such as renewable energy incentives (Narassimhan et al. 2018; Somanathan et al. 2021). Meanwhile, Wang et al. (2025) find that climate policy uncertainty can amplify carbon market volatility, thus presenting a challenge to energy transition and achieving climate targets.

Other work has examined the role of trade openness in shaping emissions. While trade facilitates economic growth, it can also lead to increased carbon emissions via heightened industrial activity. Several studies suggest greater trade openness exacerbates CO₂ emissions (e.g., Antweiler, Copeland, and Taylor 2001; McAusland 2008; Ertugrul et al. 2016; Mutascu 2018). However, the impact varies across economies, with some studies finding trade openness has heterogeneous effects based on economy-specific factors, such as comparative advantages and regulatory frameworks (Dinda and Coondoo 2006). Our paper builds on this work by considering how the extent of trade openness in emerging economies impacts the effect of EPU on emissions.

In addition, several studies have noted that R&D investment in green technologies can significantly reduce emissions, supporting the transition to a low-carbon economy. Fethi and Rahuma (2019) found that increased R&D spending negatively correlated with carbon emissions. Similarly, Khan et al. (2020) demonstrated that renewable energy adoption, income levels, and

environmental innovation contributed to long-term emissions reduction. However, other studies suggest economic innovation alone is insufficient to reduce emissions, particularly in the absence of stringent environmental regulations (Wang and Zhang 2022).

These findings indicate that the impact of innovation on emissions depends on policy frameworks and the level of technological diffusion. While the literature on carbon emissions considers empirically the role of R&D and technological advancement, the link to EPU remains relatively unexplored, however. Our paper builds upon this strand of the literature by examining how R&D investment affects the impact of EPU on carbon emissions.

III. DATA AND METHODOLOGY

A. Data

We use available yearly data with an unbalanced panel for 13 EMEs spanning from 1990 to 2023.¹ The core dependent variable, CO₂ emissions, is measured as consumption-based emissions in metric tons per capita, sourced from the Global Carbon Budget. To capture EPU, we employ the widely cited Economic Policy Uncertainty Index developed by Baker, Bloom, and Davis (2016), which quantifies uncertainty using textual analysis of over 2,000 newspapers. Following standard practice (Gulen and Ion 2016; Attig et al. 2021), monthly EPU values are averaged to construct an annualized index.

To account for potential heterogeneity in the effects of EPU on carbon emissions, we incorporate several fundamental and institutional characteristics. Specifically, we control for energy consumption (kilogram of oil equivalent per capita), energy intensity (energy use per unit of gross domestic product [GDP]), renewable energy consumption (share of total final energy consumption), trade openness (exports plus imports as a percentage of GDP), and R&D expenditure (percentage of GDP). These data are obtained from the World Bank's World Development Indicators database.

Additionally, we incorporate institutional and financial characteristics, including political stability, measured using the Political Stability and Absence of Violence/Terrorism Index from the World Bank's Worldwide Governance Indicators database; financial development, captured by the Financial Development Index from the International Monetary Fund (IMF), which assesses

¹ The sample of EMEs includes Brazil; Chile; the People's Republic of China (PRC); Colombia; Croatia; Greece; Hong Kong, China; India; the Republic of Korea; Mexico; Pakistan; Russia; and Singapore. The data starting year for a specific variable varies across economies depending on data availability and reliability.

the depth, access, and efficiency of financial institutions and markets; and climate vulnerability, measured using the Notre Dame Global Adaptation Initiative (ND-GAIN) vulnerability index.

Finally, we incorporate control variables for domestic fundamentals: we include the real GDP growth rate as a measure of domestic output growth and year-on-year changes in the consumer price index as an indicator of inflation.

B. Econometric Methodology

Following the framework proposed by Jordà (2005), we use the panel local projection to estimate the model and calculate impulse responses to the change in EPU. The baseline model can be given as follows:

$$y_{i,t+h} = \alpha_i^h + \lambda_t^h + \sum_{l=0}^L \delta_l^h z_{i,t-l} + \beta^h EPU_{i,t} + \varepsilon_{i,t+h} \quad (1)$$

where $i = 1, \dots, N$ refers to the specific economy in the sample, y is the variable of interest (namely, the logged carbon emission level), $EPU_{i,t}$ is the series of the logarithm of EPU, z is a vector of control variables including lagged values for y and $EPU_{i,t}$ as well as other macroeconomic controls (e.g., real GDP growth, inflation, and energy consumption per capita), and δ_l^h is a vector of coefficients associated with the lags of z . Specifically, we set $L = 1$, therefore we include 1 year of lagged values of z . The coefficient β^h measures the impact of a 1-standard deviation change in EPU on y at horizon h . Thus, one constructs the impulse responses as a sequence of the β^h estimated in a series of separate regressions for each horizon h . α_i denotes economy-specific fixed effects, controlling for the time-invariant characteristics of the economy. λ_t^h represents the time fixed effects.² Finally, $\varepsilon_{i,t+h}$ denotes a mean zero error term capturing other shocks. We cluster our standard errors at the economy level.

We can further adapt the local projection framework to allow for nonlinearities in the specifications that are associated with economy-specific fundamental or institutional characteristics. To explore the heterogeneous impacts of EPU on carbon emissions, we introduce a dummy variable, $I_{i,t-1}^m$, which identifies the group of economies of interest. Specifically, it takes the value of 1 if at time $t - 1$ the economy's fundamental or institutional characteristic falls within a particular level $m \in M$ of the characteristic distribution, defining the economy's group. Accordingly, η_m^h represents the group-specific intercept term associated with regime m ,

² The time fixed effects also control for structural breaks resulting from the coronavirus disease pandemic.

capturing differences in average outcomes across groups defined by the underlying characteristic. Following Cloyne et al. (2023), we specify our extended model as:

$$y_{i,t+h} = \alpha_i^h + \lambda_t^h + \sum_{l=0}^L \delta_l^h z_{i,t-l} + \sum_{m=1}^M I_{i,t-1}^m \cdot \beta_m^h EPU_{i,t} + \sum_{m=1}^M I_{i,t-1}^m \cdot \eta_m^h + u_{i,t+h} \quad (2)$$

where the notation follows that of in eq. (1).

IV. EMPIRICAL RESULTS

A. Baseline Results

Figure 1 shows the estimated impulse responses of carbon emissions to EPU, based on the linear model of eq. (1). The solid line represents the estimated impulse responses over the following 4 years to a 1-standard deviation increase in EPU. The shaded area represents 68% (dark) and 90% (light) confidence bands. The impulse responses indicate that carbon emissions increase after an increase in EPU. Following an increase in EPU, the level of carbon emission increases with a maximum impact of 0.03%. The estimated result is in line with the literature that also shows a positive impact of EPU on carbon emissions (e.g., Benlemlih and Yavaş 2024; Chiou et al. 2025).³

B. Robustness

In this section, we show that our baseline result—that a higher EPU persistently increases the level of carbon emissions—is robust across a range of additional exercises and specifications.

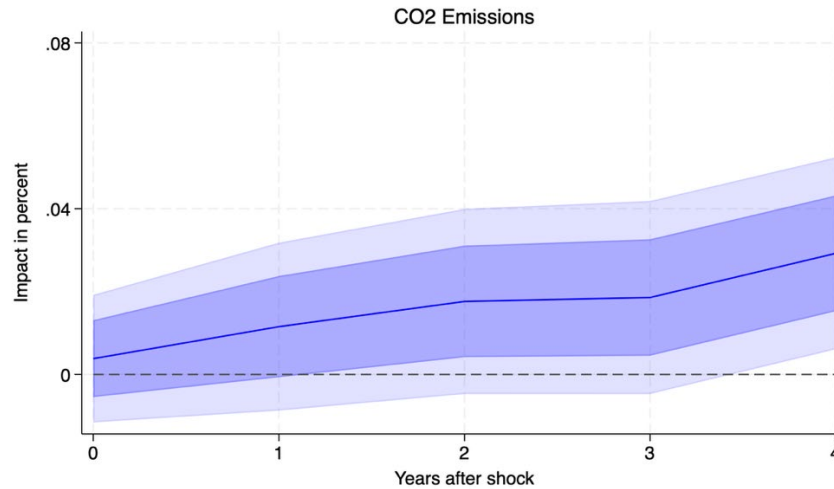
1. Adding the Quadratic Term of Gross Domestic Product Growth

To address potential nonlinearities in the relationship between economic development and carbon emissions—a core postulate of the Environmental Kuznets Curve (EKC) hypothesis (Grossman

³ Concerns around the potential downward bias in standard error estimates owing to the small cross-sectional dimension are mitigated by implementing a block bootstrap procedure with 1,000 replications to obtain inference that is more robust to the small-sample cross-sectional structure. The bootstrap results are consistent with our baseline findings (see Figure A1 in the Appendix). In addition, we implement the Split-Panel Jackknife (SPJ) estimator (Dhaene and Jochmans 2015; Chudik, Pesaran, and Yang 2018) to examine the robustness of our baseline to potential Nickell bias (Mei, Sheng, and Shi 2026). The SPJ estimates confirm a positive response of emissions to EPU shocks. While the baseline specification exhibits a gradually increasing dynamic profile, the SPJ estimates are flatter over time, with closer alignment in magnitudes from around 3 years after the shock (see Figure A2 in the Appendix). Importantly, both approaches yield the same qualitative conclusion regarding the sign and persistence of the effect. Given the relatively long time dimension of the panel ($T=34$), Nickell bias (of order $1/T$) is expected to be limited. We therefore retain the baseline specification as our main benchmark and report the SPJ estimates as a robustness check.

and Krueger 1991, Shafik 1994)—we augment our baseline model by including a quadratic term of GDP growth. This test ensures our estimates of EPU on emissions are not confounded by omitted curvature in the growth–environment nexus, and the results (Figure 2) indicate that our benchmark estimates are qualitatively robust.

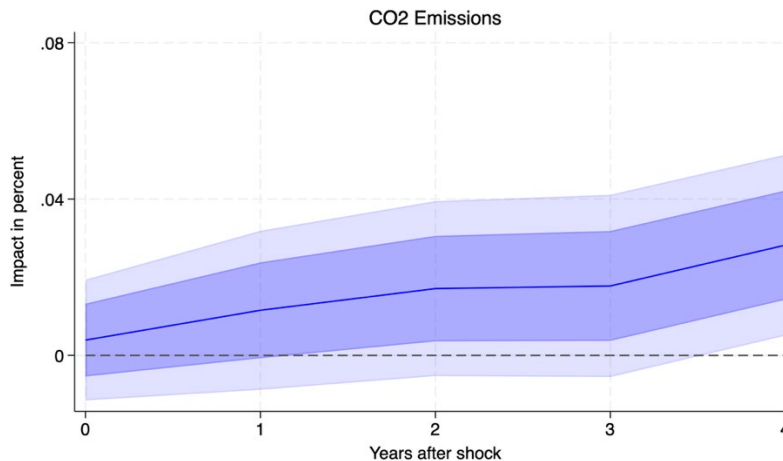
Figure 1: Dynamic Response of Carbon Emissions to Economic Policy Uncertainty



Notes: The figure plots the impulse responses of carbon emissions to a 1-standard deviation increase in economic policy uncertainty. Shaded areas represent 68% and 90% confidence bands.

Source: Authors' calculations.

Figure 2: Dynamic Response of Carbon Emissions to Economic Policy Uncertainty, Adding the Quadratic Term of Gross Domestic Product Growth



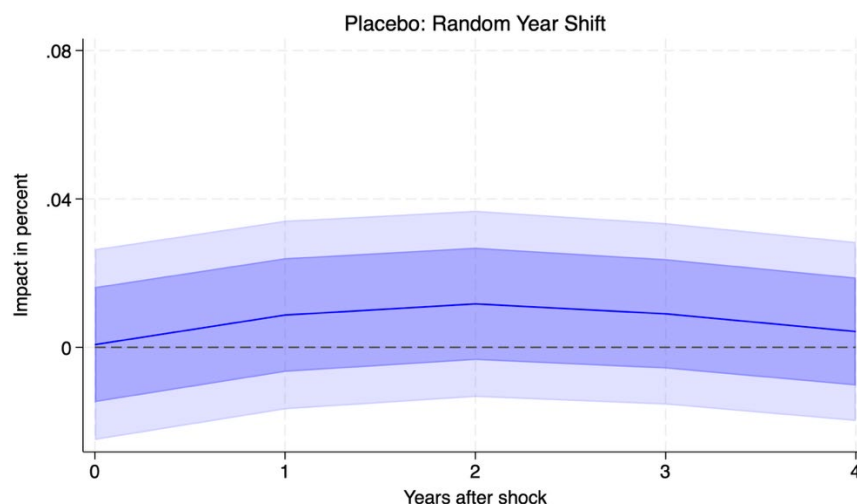
Notes: The figure plots the impulse responses of carbon emissions to a 1-standard deviation increase in economic policy uncertainty, adding the quadratic term of gross domestic product growth as additional controls. Shaded areas represent 68% and 90% confidence bands.

Source: Authors' calculations.

2. Placebo Test

To assess whether our baseline estimates merely capture spurious time-series co-movement rather than the effect of EPU shocks, we conduct a placebo test. Specifically, for each economy, we randomly shift the timing of EPU changes by an integer number of years drawn uniformly from the set $\{-2, -1, \dots, +3\}$. This procedure generates placebo EPU shocks that preserve the within-economy persistence of EPU while breaking their temporal alignment with actual economic and emissions dynamics. We then re-estimate our local projection model using these placebo shocks. As Figure 3 shows, the impulse responses to placebo shocks are close to zero and statistically insignificant across all horizons, suggesting our baseline results are unlikely to be driven solely by time-series co-movement.

Figure 3: Placebo Test



Notes: The figure plots the impulse responses of carbon emissions to a placebo shock, constructed by randomly shifting the timing of economic policy uncertainty within each economy. Shaded areas represent 68% and 90% confidence bands.

Source: Authors' calculations.

V. EXTENSIONS

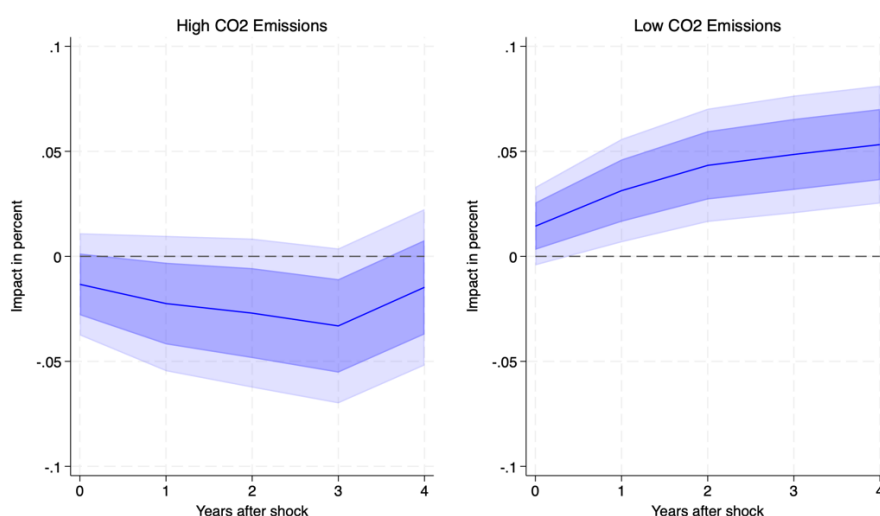
A. Impulse Responses Conditioning on Initial Carbon Emission Level

So far, we have examined the impact of EPU on carbon emissions without considering the initial carbon emission level. To investigate how the ex-ante level of carbon emissions shapes the response to changes in EPU, we focus on the differential impact of EPU on EMEs with varying initial levels of carbon emissions. Specifically, we divide our sample into two groups: high and low initial levels of carbon emissions. Specifically, we divide our sample into two groups: high and low initial carbon emission levels, using the median of our sample (4.675 metric tons per capita) as

the threshold. Figure 4 illustrates the dynamic response of carbon emissions for EMEs with low and high initial levels of carbon emissions. The estimates show that, in the low carbon emission group, the positive impact of EPU on carbon emissions is notably stronger than in the baseline case, with a significant 0.05% increase in carbon emissions occurring around 4 years after the shock. In contrast, high-carbon emission economies experience a significant decrease in carbon emissions following an increase in EPU.

These findings highlight the divergent impact of EPU on carbon emissions in high- and low-emission economies. In high-emission economies, an increase in policy uncertainty is associated with a reduction in carbon emissions, likely because of economic slowdowns that lead to lower industrial production and energy demand. This is consistent with the literature suggesting that uncertainty can defer investments and reduce output in energy-intensive sectors (Stern 2011). On the other hand, in low-emission economies, EPU seems to drive an increase in carbon emissions, potentially because these economies prioritize short-term economic growth over environmental concerns. Governments in these economies may expand carbon-intensive industries or delay the transition to cleaner technologies during periods of uncertainty. Furthermore, these economies may rely more heavily on high-emission sectors to stimulate economic activity when faced with heightened policy uncertainty.

Figure 4: Dynamic Response of Carbon Emissions to Economic Policy Uncertainty, Conditioning on Initial Carbon Emission Level



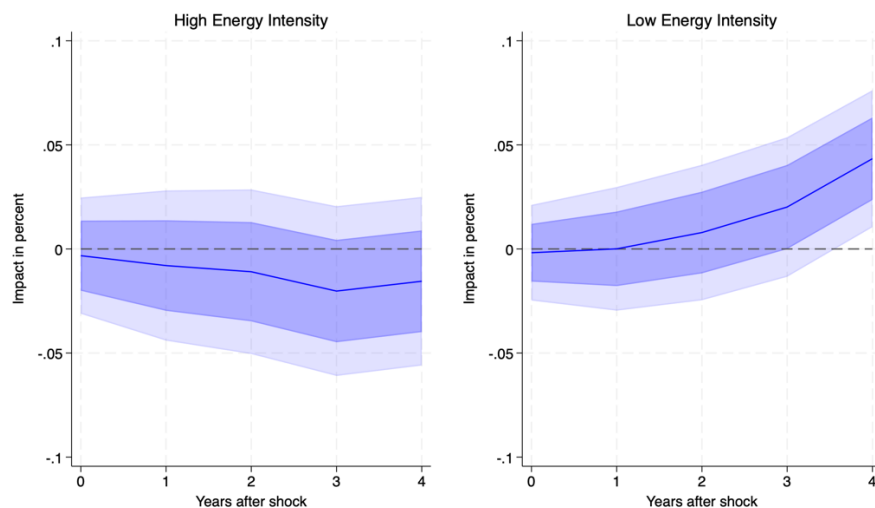
Notes: The figure plots the impulse responses of carbon emissions to a 1-standard deviation increase in economic policy uncertainty. Shaded areas represent 68% and 90% confidence bands.

Source: Authors' calculations.

B. Impulse Responses Conditioning on Energy Intensity

To further explore the relationship between EPU and carbon emissions, we analyze how energy intensity influences this dynamic. Energy intensity, which measures the energy inefficiency of an economy, is calculated as units of energy per unit of GDP. A higher value of energy intensity indicates a higher cost of converting energy into GDP. Figure 5 reports the impulse responses of carbon emissions for EMEs with low and high energy intensity, using the median value of the sample (1.351) as the threshold. Our results show low-energy intensity economies tend to experience an increase in carbon emissions in response to rising EPU. This could be because these economies may rely more on energy-intensive sectors during uncertain times, potentially leading to higher carbon emissions as they attempt to stabilize their economies (Antweiler, Copeland, and Taylor 2001). In contrast, high-energy intensity economies exhibit insignificant responses of carbon emissions to EPU shocks. This muted effect suggests that, although uncertainty may dampen economic activity, the strong dependence on energy-intensive production and structural rigidities limits the extent to which emissions adjust in the short run (Bloom 2009, 2014).

Figure 5: Dynamic Response of Carbon Emissions to Economic Policy Uncertainty, Conditioning on Energy Intensity



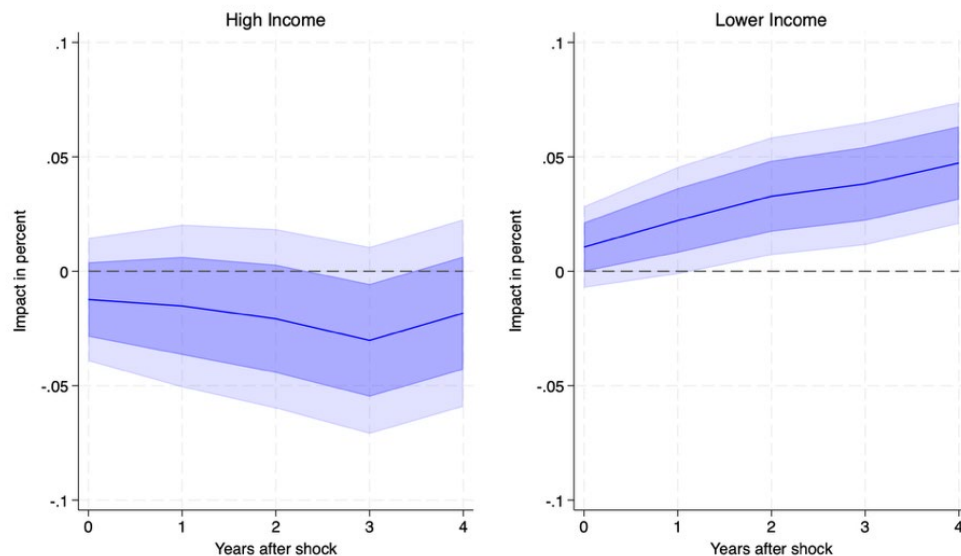
Notes: The figure plots the impulse responses of carbon emissions to a 1-standard deviation increase in economic policy uncertainty. Shaded areas represent 68% and 90% confidence bands.

Source: Authors' calculations.

C. Impulse Responses Conditioning on Income Levels

Next, we examine how the income level of an economy affects its response to an increase in EPU. We divide our sample of EMEs into high-income and low-income economies, following the World Bank income classification.⁴ Figure 6 presents the impulse responses of carbon emissions to a rise in EPU for both income groups. We show that, in high-income EMEs, the impact of EPU on carbon emissions is slightly negative or statistically insignificant. This could be attributed to the greater resilience of high-income economies, which often have more diversified industries, robust institutional frameworks, and a higher capacity to adapt to economic shocks. As a result, these economies tend to experience a quicker rebound in emissions after periods of uncertainty (Grossman and Krueger 1991). In contrast, low-income EMEs exhibit positive responses. This can be explained by structural vulnerabilities in lower-income economies, including a heavier reliance on energy-intensive sectors, underdeveloped energy infrastructure, and limited capacity to implement effective climate policies during times of economic uncertainty (Shafik 1994).

Figure 6: Dynamic Response of Carbon Emissions to Economic Policy Uncertainty, Conditioning on Income Level



Notes: The figure plots the impulse responses of carbon emissions to a 1-standard deviation increase in economic policy uncertainty. Shaded areas represent 68% and 90% confidence bands.

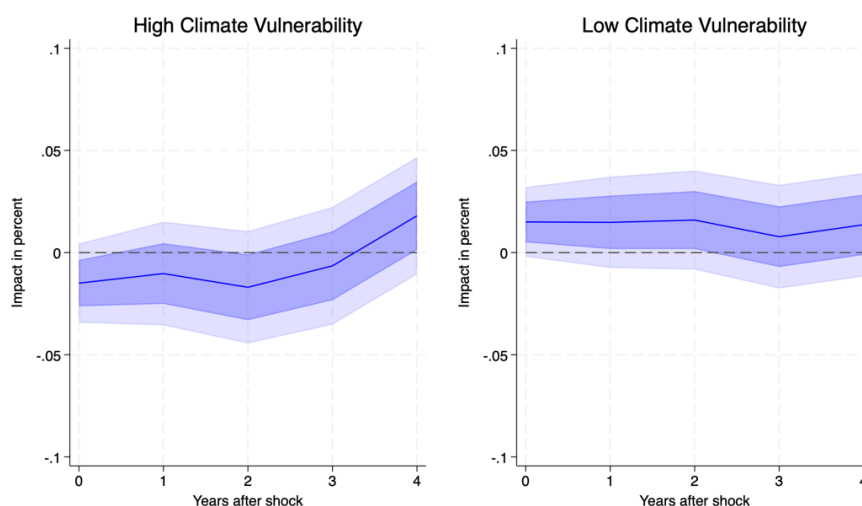
Source: Authors' calculations.

⁴ We define lower-income economies as the combination of low, lower-middle income, and upper-middle income economies in our EME sample, to include Brazil, the PRC, Colombia, Croatia, India, Mexico, and Russia.

D. Impulse Responses Conditioning on Climate Risk Vulnerability

We next examine whether an economy's climate risk vulnerability influences the impact of EPU on carbon emissions. Figure 7 presents the impulse responses of carbon emissions depending on climate risk vulnerability, using the median value of the sample (0.379) as the threshold to classify economies as either high or low climate risk vulnerability. We find that high-climate risk vulnerability EMEs experience a slightly negative or insignificant response in carbon emissions to rising EPU, reflecting stronger incentives to maintain environmental commitments in economies that are more exposed to climate-related risks. In particular, greater exposure to physical and transition risks may induce more persistent climate policy efforts and discourage carbon-intensive expansion even during periods of uncertainty (Meckling, Sterner, and Wagner 2017; Best, Burke, and Jotzo 2020). Conversely, low-climate risk vulnerability EMEs exhibit a short-term positive response to rising EPU. This can be attributed to these economies' relatively lower perceived risks from climate change, prompting them to prioritize economic stability and growth over environmental concerns, often expanding carbon-intensive industries (Antweiler, Copeland, and Taylor 2001; Benlemlih and Yavaş 2024).

Figure 7: Dynamic Response of Carbon Emissions to Economic Policy Uncertainty, Conditioning on Climate Vulnerability



Notes: The figure plots the impulse responses of carbon emissions to a 1-standard deviation increase in economic policy uncertainty. Shaded areas represent 68% and 90% confidence bands.

Source: Authors' calculations.

E. Impulse Responses Conditioning on Implementing Carbon Pricing Instruments

We also explore whether the implementation of carbon pricing instruments influences how carbon emissions respond to EPU. To this end, we construct a time-varying dummy variable that indicates whether a given economy has a carbon pricing mechanism in place (i.e., carbon taxes or emissions trading schemes) at a specific point in time.⁵ Figure 8 presents the heterogeneous responses of carbon emissions, comparing EMEs with and without carbon pricing mechanisms. The estimates show EMEs with established carbon pricing mechanisms respond insignificantly in carbon emissions to rising EPU. This is consistent with the notion that these economies are more committed to long-term climate goals, even in the face of economic fluctuations (Haites 2018). Carbon pricing likely acts as a stabilizing force in these economies, ensuring environmental policies remain effective despite policy uncertainty. In contrast, EMEs without carbon pricing instruments experience a positive response, with emissions increasing in response to EPU. This likely owes to the absence of market-based incentives for emissions reductions, leaving these economies more susceptible to the short-term economic pressures and volatility induced by policy uncertainty (Best, Burke, and Jotzo 2020).

These findings suggest the presence of carbon pricing mechanisms can help mitigate the impact of EPU on carbon emissions, providing a stabilizing effect that supports long-term environmental objectives.

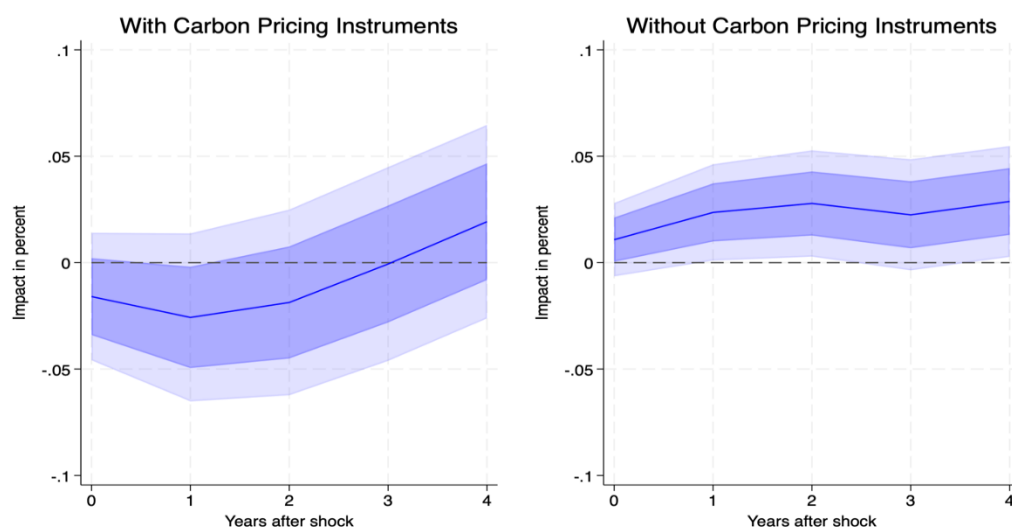
F. Impulse Responses Conditioning on Political Stability

We further explore the role of political stability in shaping the impact of EPU on carbon emissions in EMEs. Figure 9 presents the heterogeneous effects of EPU shocks on carbon emissions based on political stability, using the median value of our sample (-0.220) to categorize economies into high and low political stability. The results indicate EMEs with high political stability experience a negative response to rising EPU, suggesting stable political environments allow for more effective and consistent implementation of environmental policies. Strong governance structures and long-term policy commitments enable these economies to mitigate the adverse effects of uncertainty, leading to reductions in carbon emissions (e.g., Dinda and Coondoo 2006, Wang and Zhang 2022). In contrast, EMEs with low political stability show a positive response, with carbon emissions increasing in response to EPU. This can be attributed to policy fragmentation, weaker

⁵ Specifically, the variable takes a value of 1 if a country implements any form of carbon pricing (e.g., a carbon tax or an emissions trading system) and 0 otherwise. The carbon pricing information is compiled from publicly available sources provided by the Organisation for Economic Co-operation and Development (OECD) and the IMF, which document the adoption and timing of carbon pricing policies across economies.

institutional capacity, and the prioritization of short-term economic goals over environmental concerns in politically unstable settings (Bloom 2009, 2014).

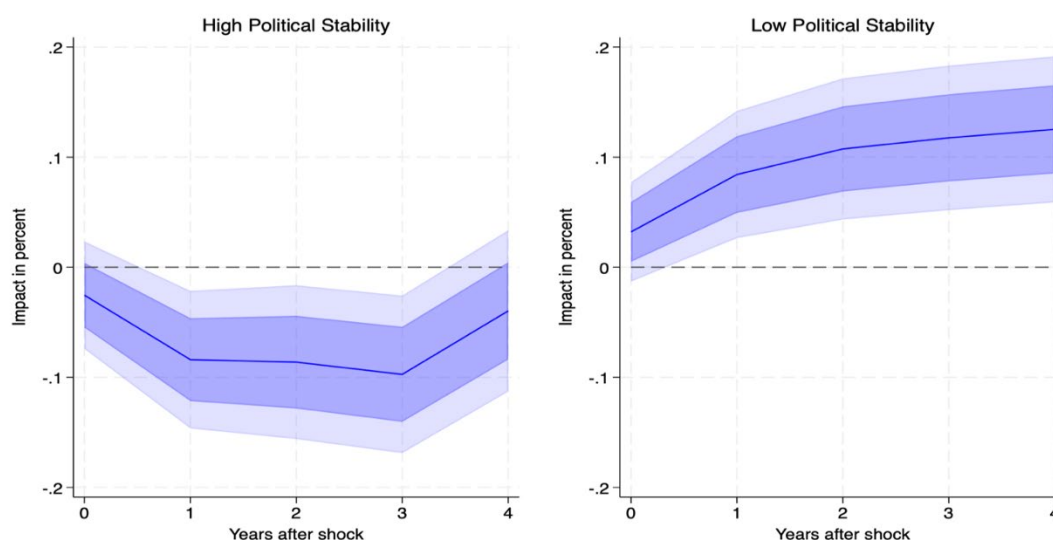
Figure 8: Dynamic Response of Carbon Emissions to Economic Policy Uncertainty, Conditioning on Implementing Carbon Pricing Instruments



Notes: The figure plots the impulse responses of carbon emissions to a 1-standard deviation increase in economic policy uncertainty. Shaded areas represent 68% and 90% confidence bands.

Source: Authors' calculations.

Figure 9: Dynamic Response of Carbon Emissions to Economic Policy Uncertainty, Conditioning on Political Stability



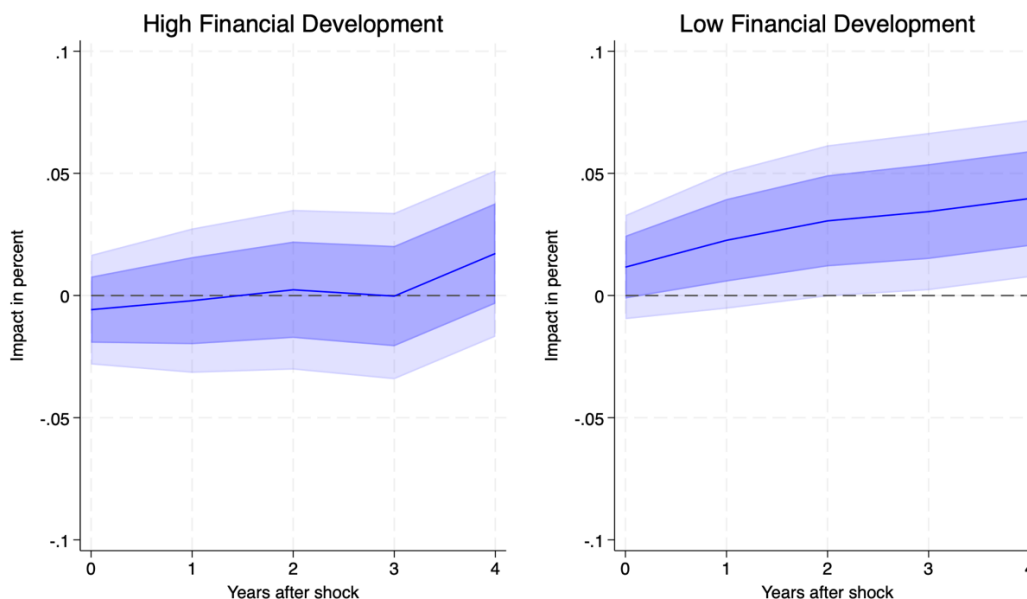
Notes: The figure plots the impulse responses of carbon emissions to a 1-standard deviation increase in economic policy uncertainty. Shaded areas represent 68% and 90% confidence bands.

Source: Authors' calculations.

G. Impulse Responses Conditioning on Financial Development

Next, we investigate whether financial development influences the response of carbon emissions to economic policy uncertainty. Specifically, we categorize EMEs into high and low financial development groups based on the median value of the sample (0.470). Figure 10 illustrates the heterogeneous responses of carbon emissions to an increase in EPU, conditioned on financial development. The results show EMEs with high financial development experience a muted effect of EPU on carbon emissions, which suggests well-developed financial systems contribute to economic resilience, enabling continued investment in low-carbon technologies and facilitating sustainable energy transitions even during periods of uncertainty (e.g., Renzhi and Baek 2020). On the other hand, EMEs with low financial development show amplified and persistent effects, attributed to limited access to financing for green investments, weaker institutional frameworks, and heightened economic fragility in these economies (Fethi and Rahuma 2019).

Figure 10: Dynamic Response of Carbon Emissions to Economic Policy Uncertainty, Conditioning on Financial Development



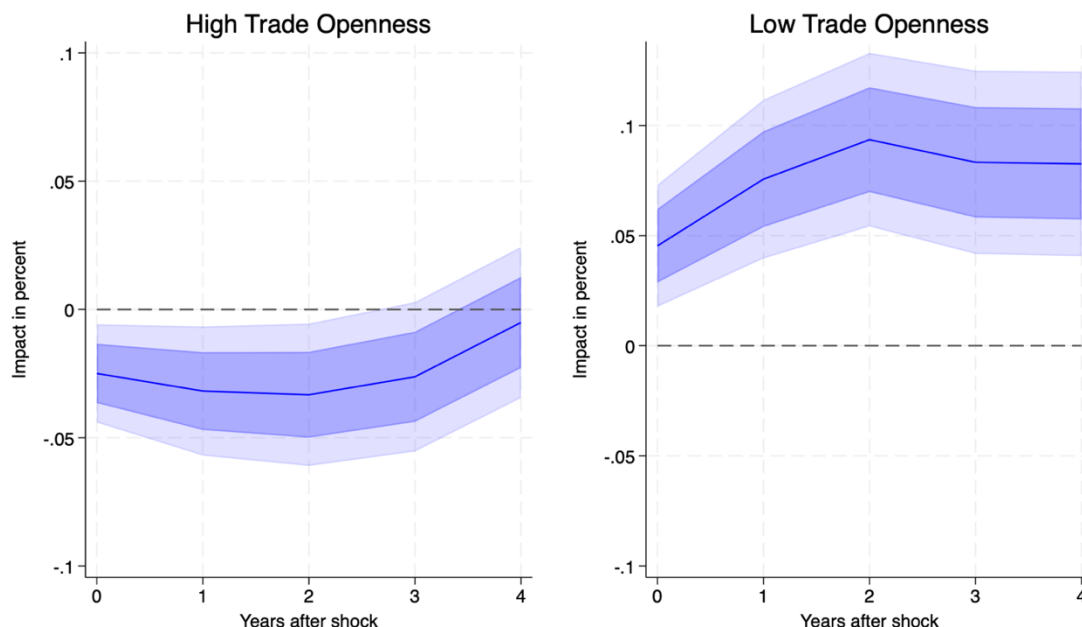
Notes: The figure plots the impulse responses of carbon emissions to a 1-standard deviation increase in economic policy uncertainty. Shaded areas represent 68% and 90% confidence bands.

Source: Authors' calculations.

H. Impulse Responses Conditioning on Trade Openness

We further examine whether trade openness plays a significant role in shaping the effects of EPU on carbon emissions in EMEs. Figure 11 presents the heterogeneous responses of carbon emissions to an increase in EPU, conditional on trade openness, where economies are divided into high- and low-openness groups using the sample median (52.041) as the threshold. The results show high-trade openness EMEs exhibit a negative response to EPU, suggesting greater integration into global markets contributes to more stable and effective environmental policies. In these economies, stronger trade linkages may incentivize adherence to international climate commitments, while access to foreign capital and technology facilitates investment in cleaner energy sources even during periods of uncertainty (e.g., Copeland and Taylor 2004). Conversely, low-trade openness EMEs show a positive response, with carbon emissions rising amid economic policy uncertainty. This can be attributed to a greater reliance on domestic markets, where policy uncertainty may lead to increased dependence on carbon-intensive industries to sustain economic activity (McAusland 2008). Overall, our findings highlight how external economic integration can help buffer against the negative environmental consequences of policy uncertainty.

Figure 11: Dynamic Response of Carbon Emissions to Economic Policy Uncertainty, Conditioning on Trade Openness



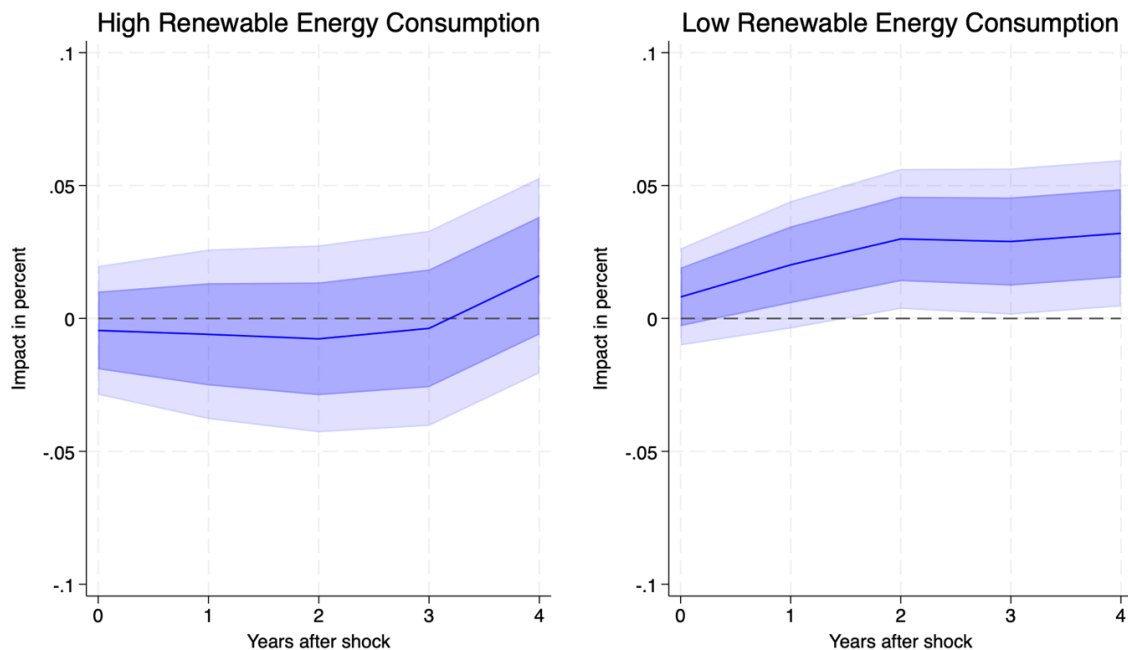
Notes: The figure plots the impulse responses of carbon emissions to a 1-standard deviation increase in economic policy uncertainty. Shaded areas represent 68% and 90% confidence bands.

Source: Authors' calculations.

I. Impulse Responses Conditioning on Share of Renewable Energy Consumption

Next, we investigate the role of renewable energy consumption in shaping the impact of EPU on carbon emissions. We classify EMEs into high and low renewable energy consumption groups based on their share of renewables in total energy consumption, using the median value of our sample (20.15) as the threshold. Figure 12 shows economies with a high reliance on renewable energy respond insignificantly to EPU changes, suggesting a well-established renewable energy infrastructure provides resilience against economic fluctuations, ensuring emissions remain stable even in the face of policy uncertainty (Stern 2011, Khan et al. 2020). In contrast, EMEs with low renewable energy consumption exhibit a persistent and positive response, reflecting the discouraging effect of uncertainty on long-term green investment, which reinforces dependence on existing carbon-intensive infrastructure. Our findings align with evidence that uncertainty can depress investment in new technologies and slow structural transformation (Bloom 2009, 2014), particularly in economies with less developed clean energy sectors.

Figure 12: Dynamic Response of Carbon Emissions to Economic Policy Uncertainty, Conditioning on Share of Renewable Energy



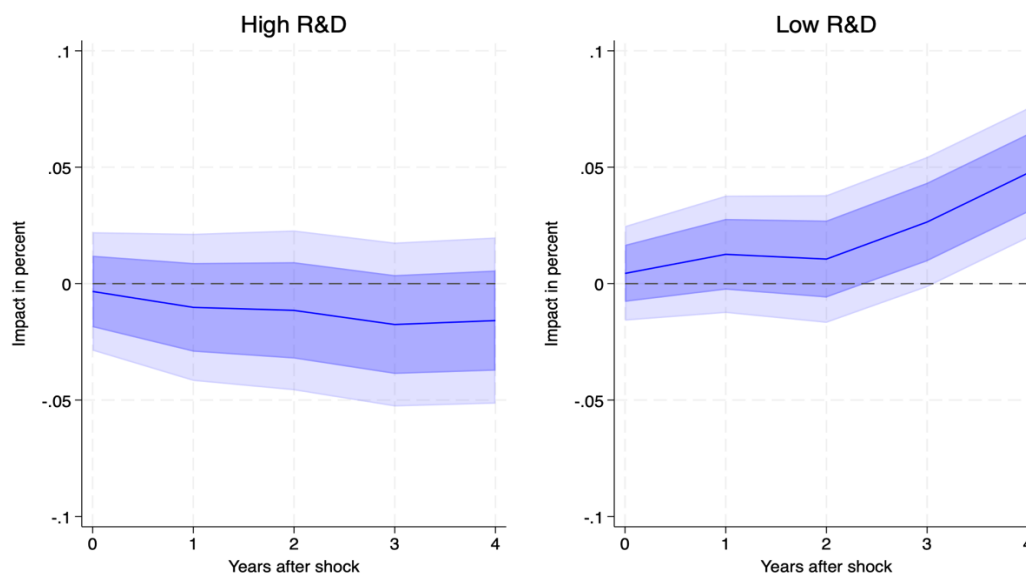
Notes: The figure plots the impulse responses of carbon emissions to a 1-standard deviation increase in economic policy uncertainty. Shaded areas represent 68% and 90% confidence bands.

Source: Authors' calculations.

J. Impulse Responses Conditioning on Share of Research and Development Expenditure

Our final experiment examines the role of R&D intensity in shaping the impact of EPU on carbon emissions. We categorize EMEs into high and low R&D expenditure groups based on their share of GDP allocated to R&D, using the median value of our sample (0.835) as the threshold. Figure 13 exhibits that, in high-R&D economies, the response of carbon emissions to EPU shocks is statistically insignificant, suggesting economies with strong innovation capacity and technological advancements are better equipped to maintain their green transition efforts despite periods of uncertainty (e.g., Zhang et al. 2025). Well-developed R&D sectors likely support continued improvements in energy efficiency and low-carbon technologies, mitigating the adverse environmental effects of economic uncertainty (Khan et al. 2020). In contrast, low-R&D economies experience a significant positive response, stemming from weaker technological capabilities, reduced investment in green innovation, and higher dependence on conventional energy sources, which become more entrenched during uncertain periods (Gulen and Ion 2016). Our findings underscore the role of R&D investment in enhancing economic resilience and sustaining environmental commitments even in uncertain policy environments.

Figure 13: Dynamic Response of Carbon Emissions to Economic Policy Uncertainty, Conditioning on Share of Research and Development Expenditure



R&D = research and development.

Notes: The figure plots the impulse responses of carbon emissions to a 1-standard deviation increase in economic policy uncertainty. Shaded areas represent 68% and 90% confidence bands.

Source: Authors' calculations.

VI. CONCLUSIONS AND POLICY IMPLICATIONS

In this paper, we empirically examine the dynamic effects of EPU on carbon emissions in EMEs using panel local projections. Our findings indicate that an increase in EPU leads to a significant rise in carbon emissions. Subsequent analysis reveals that various structural and institutional characteristics influence the magnitude and persistence of this effect. Specifically, economies with lower initial carbon emissions, higher energy intensity, implemented carbon pricing mechanisms, greater financial development, higher trade openness, stronger political stability, greater reliance on renewable energy, and higher R&D intensity are better positioned to mitigate the adverse impact of EPU. Additionally, we find high-income EMEs experience only a weak impact in emissions as a result of EPU, whereas lower-income EMEs exhibit a more persistent response. Furthermore, the impact of EPU is significantly positive in economies with low climate vulnerability, whereas high-climate vulnerability EMEs display a more muted response, possibly because of stronger climate adaptation policies and regulatory frameworks.

Our findings have implications for policymakers aiming to enhance economic resilience to mitigate the adverse environmental consequences of EPU. Strengthening institutional frameworks, promoting financial development, and fostering trade openness can help stabilize carbon emissions in uncertain economic environments. Moreover, expanding carbon pricing mechanisms and increasing investments in renewable energy and R&D are crucial in reducing the sensitivity of emissions to policy uncertainty. For lower-income EMEs, targeted policies that enhance climate adaptation capacity and provide financial support for green investments can help curb the prolonged impact of uncertainty on emissions. Additionally, economies with lower climate vulnerability should proactively integrate environmental policies to prevent emissions surges during periods of economic instability. Overall, policymakers should prioritize long-term sustainability strategies that reinforce economic and institutional resilience, ensuring environmental objectives remain on track despite fluctuations in policy uncertainty.

APPENDIX

Table A1: Overview of Variables Used in the Empirical Analysis

Variable	Source	Definition
Economic policy uncertainty (log)	Baker, Bloom, and Davis (2016)	Logarithm of Economic Policy Uncertainty Index developed by Baker, Bloom, and Davis (2016), which quantifies uncertainty using textual analysis of over 2,000 newspapers. The monthly EPU values are averaged to construct an annualized index.
Carbon dioxide emission (log)	Global Carbon Budget	Logarithm of consumption-based CO ₂ emissions (metric tons per capita).
Real gross domestic product growth (%)	World Bank	Real GDP growth rate.
Inflation (%)	World Bank	Year-on-year changes in the consumer price index.
Energy consumption (log)	World Bank	Logarithm of primary energy consumption (kilograms of oil equivalent per capita).
Energy intensity (log)	World Bank	Logarithm of primary energy consumption per unit of GDP (measured at purchasing power parity).
Climate vulnerability (index)	ND-GAIN	ND-GAIN Vulnerability Index, which summarizes an economy's vulnerability to climate change.
Trade openness (%)	World Bank	Sum of exports and imports of goods and services as a share of GDP.
Renewable energy intensity (%)	World Bank	Share of renewable energy in total final energy consumption.
Research and development intensity (%)	World Bank	R&D expenditure as a share of GDP.
Political stability (index)	World Bank	Political Stability and Absence of Violence/Terrorism Index, which measures perceptions of the likelihood a government will be destabilized or overthrown by unconstitutional or violent means, including terrorism.
Financial development (index)	IMF	Financial Development Index, which measures the depth, access, and efficiency of financial institutions and markets.
Carbon pricing implementation (dummy)	IMF and OECD	Dummy variable that equals 1 if an economy implements any form of carbon pricing (e.g., a carbon tax or an emissions trading system) and 0 otherwise. Information on carbon pricing is compiled from publicly available sources provided by the OECD and the IMF, which document the adoption and timing of carbon pricing policies across economies.

Source: Authors' compilation.

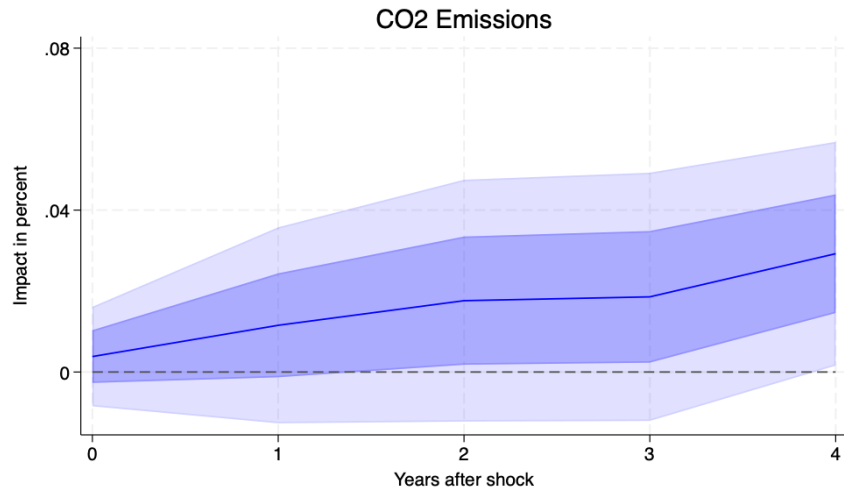
Table A2: Summary Statistics

Variable	Mean	Median	Standard Deviation	Min	Max	Skewness	Kurtosis
Economic policy uncertainty (index)	129.183	102.830	85.640	27.001	588.373	2.520	11.444
Carbon dioxide emission (metric tons per capita)	7.342	4.675	7.729	0.587	47.559	2.138	8.245
Real gross domestic product growth (%)	3.669	3.936	4.461	-21.089	14.520	-1.109	6.320
Inflation (%)	35.052	4.367	220.142	-4.009	2947.733	9.659	105.860
Energy consumption (log)	7.364	7.431	0.796	5.868	8.905	-0.191	2.146
Energy intensity (log)	1.389	1.351	0.468	0.157	2.490	0.081	2.992
Climate vulnerability (index)	0.397	0.379	0.059	0.322	0.546	1.098	3.111
Trade openness (%)	94.611	52.041	108.190	15.156	442.620	1.948	5.277
Renewable energy intensity (%)	21.234	20.150	17.598	0.000	58.100	0.268	1.677
Research and development intensity (%)	1.093	0.835	0.899	0.109	4.930	1.865	7.005
Political stability (index)	-0.234	-0.220	0.979	-2.810	1.599	-0.396	2.615
Financial development (index)	0.478	0.470	0.182	0.000	0.853	0.203	2.448
Carbon pricing implementation (dummy)	0.238	0.000	0.426	0.000	1.000	1.233	2.521

Note: The table shows summary statistics for the main variables used in the empirical analysis.

Source: Authors' calculations.

Figure A1: Dynamic Response of Carbon Emissions to Economic Policy Uncertainty, Bootstrap-Based



Notes: The figure plots the impulse responses of carbon emissions to a 1-standard deviation increase in economic policy uncertainty. Inference is based on a block bootstrap procedure with 1,000 replications. Shaded areas represent 68% and 90% confidence bands.

Source: Authors' calculations.

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Economic Policy Uncertainty and Carbon Emissions in Emerging Markets

This paper examines how economic policy uncertainty (EPU) affects carbon emissions in 13 emerging market economies from 1990 to 2023. Results show that higher EPU increases emissions, though effects vary by structural and institutional factors. Economies with stronger institutions, cleaner energy, and higher development levels exhibit weaker responses. Lower-income economies face more pronounced effects, while climate vulnerability also shapes outcomes. The findings highlight how policy and structural conditions influence emissions under uncertainty and inform strategies to mitigate environmental risks.

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